DDL Commands using CREATE, ALTER, TRUNCATE, and DROP

```
mysql> alter table student modify student_name varchar(100);
ERROR 1054 (42S22): Unknown column 'student_name' in 'student'
mysql> alter table s modify student_name varchar(100);
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> alter table student rename column department to dept;
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> alter table student drop column email;
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> insert into student values
    -> ng
    -> hg;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to
 use near 'ng
hg' at line 2
mysql> insert into student values(101,'priya',19,'CSE'),(112,'arun',17,'ECE');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> truncate table student;
Query OK, 0 rows affected (0.04 sec)

mysql> drop table student;
Query OK, 0 rows affected (0.02 sec)
```

DDL Commands with constraints – PRIMARY,FOREIGN KEY,UNIQUE,CHECK,DEFAULT,NOT NULL

```
mysql> create table department(dept_id int primary key,dept_name varchar(50) unique);
ERROR 1050 (42S01): Table 'department' already exists
mysql> create table depart(dept_id int primary key,dept_name varchar(50) unique);
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> create table stud(student_id int primary key,student_name varchar(50) not null,age int check(age <= 17),dept_id int,email varchar(100) unique,city va
rchar(30) default'chennai',foreign key(dept_id) references department(dept_id));
Query OK, 0 rows affected (0.02 sec)

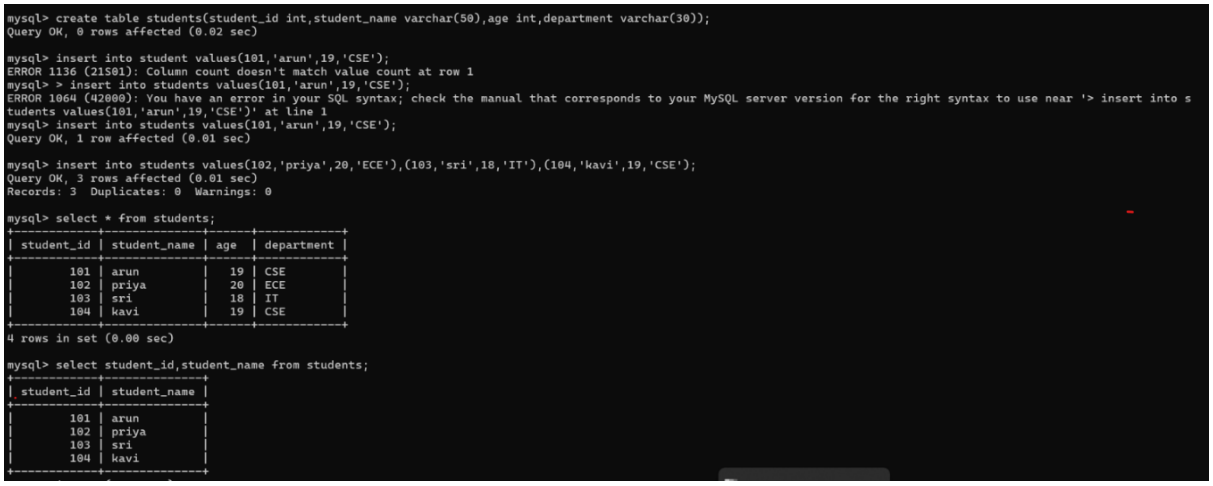
mysql> insert into dept values(1,'CSE'),(2,'ECE'),(3,'IT');
ERROR 1146 (42S02): Table 'lab.dept' doesn't exist
mysql>
mysql> mysql> insert into depart values(1,'CSE'),(2,'ECE'),(3,'IT');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'm
ysql> insert into depart values(1,'CSE'),(2,'ECE'),(3,'IT')' at line 1
mysql> insert into depart values(1,'CSE'),(2,'ECE'),(3,'IT');
Query OK, 3 rows affected (0.04 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql>  insert into student(student_id,student_name,age,dept_id,email) values(101,'arun',14,1,'arun@gmail.com'),(102,'priya',15,2,'priya@gmail.com'),(103,'k
umar',16,3,'kumar@gmail.com');
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> select * from students;
ERROR 1146 (42S02): Table 'lab.students' doesn't exist
mysql> select * from student;
+------------+--------------+------+---------+-----------------+---------+
| student_id | student_name | age  | dept_id | email           | city    |
+------------+--------------+------+---------+-----------------+---------+
|        101 | arun         |   14 |       1 | arun@gmail.com  | chennai |
|        102 | priya        |   15 |       2 | priya@gmail.com | chennai |
|        103 | kumar        |   16 |       3 | kumar@gmail.com | chennai |
+------------+--------------+------+---------+-----------------+---------+
3 rows in set (0.00 sec)

mysql> select * from department;
+---------+-----------+
| dept_id | dept_name |
+---------+-----------+
|       1 | CSE       |
|       2 | ECE       |
|       3 | IT        |
+---------+-----------+
3 rows in set (0.00 sec)
```

DML Commands using INSERT, SELECT



```
mysql> create table students(student_id int,student_name varchar(50),age int,department varchar(30));
Query OK, 0 rows affected (0.02 sec)

mysql> insert into student values(101,'arun',19,'CSE');
ERROR 1136 (21S01): Column count doesn't match value count at row 1
mysql> > insert into students values(101,'arun',19,'CSE');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '> insert into s
tudents values(101,'arun',19,'CSE')' at line 1
mysql> insert into students values(101,'arun',19,'CSE');
Query OK, 1 row affected (0.01 sec)

mysql> insert into students values(102,'priya',20,'ECE'),(103,'sri',18,'IT'),(104,'kavi',19,'CSE');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> select * from students;
+------------+--------------+------+------------+
| student_id | student_name | age  | department |
+------------+--------------+------+------------+
|        101 | arun         |   19 | CSE        |
|        102 | priya        |   20 | ECE        |
|        103 | sri          |   18 | IT         |
|        104 | kavi         |   19 | CSE        |
+------------+--------------+------+------------+
4 rows in set (0.00 sec)

mysql> select student_id,student_name from students;
+------------+--------------+
| student_id | student_name |
+------------+--------------+
|        101 | arun         |
|        102 | priya        |
|        103 | sri          |
|        104 | kavi         |
+------------+--------------+
```

```
mysql> select * from students where department = 'CSE';
+------------+--------------+------+------------+
| student_id | student_name | age  | department |
+------------+--------------+------+------------+
|        101 | arun         |   19 | CSE        |
|        104 | kavi         |   19 | CSE        |
+------------+--------------+------+------------+
2 rows in set (0.00 sec)

mysql> select * from students where age > 19;
+------------+--------------+------+------------+
| student_id | student_name | age  | department |
+------------+--------------+------+------------+
|        102 | priya        |   20 | ECE        |
+------------+--------------+------+------------+
1 row in set (0.00 sec)

mysql> select * from student order by student_name ASC;
+------------+--------------+------+---------+-----------------+---------+
| student_id | student_name | age  | dept_id | email           | city    |
+------------+--------------+------+---------+-----------------+---------+
|        101 | arun         |   14 |       1 | arun@gmail.com  | chennai |
|        103 | kumar        |   16 |       3 | kumar@gmail.com | chennai |
|        102 | priya        |   15 |       2 | priya@gmail.com | chennai |
+------------+--------------+------+---------+-----------------+---------+
3 rows in set (0.00 sec)

mysql>
```

DML Commands using UPDATE,DELETE

```
mysql> create table stdnt(student_id int primary key,student_name varchar(50),age int,department varchar(30
Query OK, 0 rows affected (0.02 sec)

mysql> insert into stdnt values(101,'arun',19,'cse'),(102,'sri',20,'ece'),(103,'priya',18,'it');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> select * from stdnt;
+------------+--------------+------+------------+
| student_id | student_name | age  | department |
+------------+--------------+------+------------+
|        101 | arun         |   19 | cse        |
|        102 | sri          |   20 | ece        |
|        103 | priya        |   18 | it         |
+------------+--------------+------+------------+
3 rows in set (0.00 sec)

mysql> update stdnt set age = 21 where student_id = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> update stdnt set department = 'it' where student_id = 102;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from students;
+------------+--------------+------+------------+
| student_id | student_name | age  | department |
+------------+--------------+------+------------+
|        101 | arun         |   19 | CSE        |
|        102 | priya        |   20 | ECE        |
|        103 | sri          |   18 | IT         |
|        104 | kavi         |   19 | CSE        |
+------------+--------------+------+------------+
4 rows in set (0.00 sec)
```

```
mysql> select * from stdnt;
+------------+--------------+------+------------+
| student_id | student_name | age  | department |
+------------+--------------+------+------------+
|        101 | arun         |   21 | cse        |
|        102 | sri          |   20 | it         |
+------------+--------------+------+------------+
2 rows in set (0.00 sec)

mysql> delete from stdnt;
Query OK, 2 rows affected (0.00 sec)
```

TCL Commands –
COMMIT,SAVEPOINT,ROLLBACK

```
mysql>  create table st(student_id int primary key,student_name varchar(50),age int);
Query OK, 0 rows affected (0.01 sec)

mysql> insert into st values(101,'arun',19),(102,'sri',20);
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> commit;
Query OK, 0 rows affected (0.00 sec)

mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)

mysql> insert into st values(103,'kumar',18);
Query OK, 1 row affected (0.00 sec)

mysql> savepoint s1;
Query OK, 0 rows affected (0.00 sec)

mysql> insert into st values(104,'divya',19);
Query OK, 1 row affected (0.00 sec)
```

```
mysql> select * from st;
+------------+--------------+------+
| student_id | student_name | age  |
+------------+--------------+------+
|        101 | arun         |   19 |
|        102 | sri          |   20 |
|        103 | kumar        |   18 |
|        104 | divya        |   19 |
+------------+--------------+------+
4 rows in set (0.00 sec)

mysql> rollback to s1;
Query OK, 0 rows affected (0.00 sec)

mysql> select * from st;
+------------+--------------+------+
| student_id | student_name | age  |
+------------+--------------+------+
|        101 | arun         |   19 |
|        102 | sri          |   20 |
|        103 | kumar        |   18 |
+------------+--------------+------+
3 rows in set (0.00 sec)

mysql> commit;
Query OK, 0 rows affected (0.01 sec)

mysql> select * from st;
+------------+--------------+------+
| student_id | student_name | age  |
+------------+--------------+------+
|        101 | arun         |   19 |
|        102 | sri          |   20 |
|        103 | kumar        |   18 |
+------------+--------------+------+
3 rows in set (0.00 sec)
```